

4.13.19

EE25BTECH11014 - Bhoomika Lokesh

Question: If the line $2x + y = k$ passes through the point which divides the line segment joining the points $(1, 1)$ and $(2, 4)$ in the ratio $3:2$, then k equals

1) $\frac{29}{5}$

2) 5

3) 6

4) $\frac{11}{5}$

Solution:

Let the point **C** divide the line segment joining **A** and **B** in the ratio $3:2$ where

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1)$$

$$\mathbf{B} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad (2)$$

By section formula,

$$(\lambda + 1)\mathbf{C} = \lambda\mathbf{B} + \mathbf{A}. \quad (3)$$

$$\text{Here } \left(\lambda = \frac{3}{2} \right) \quad (4)$$

$$\frac{5}{2} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (5)$$

$$\frac{5}{2} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (6)$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{2}{5} \begin{pmatrix} 4 \\ 7 \end{pmatrix} \quad (7)$$

$$\therefore \mathbf{C} = \begin{pmatrix} \frac{8}{5} \\ \frac{14}{5} \end{pmatrix}. \quad (8)$$

The equation of the line can be represented as:

$$\mathbf{n}^T \mathbf{x} = c_1 \quad (9)$$

$$\text{where } \mathbf{n}^T = \begin{pmatrix} 2 & 1 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \text{ and } c_1 = k. \quad (10)$$

As **C** lies on the above line,

$$\mathbf{n}^T \mathbf{C} = c_1 \quad (11)$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} \frac{8}{5} \\ \frac{14}{5} \end{pmatrix} = k \quad (12)$$

$$k = \frac{16}{5} + \frac{14}{5} = 6. \quad (13)$$

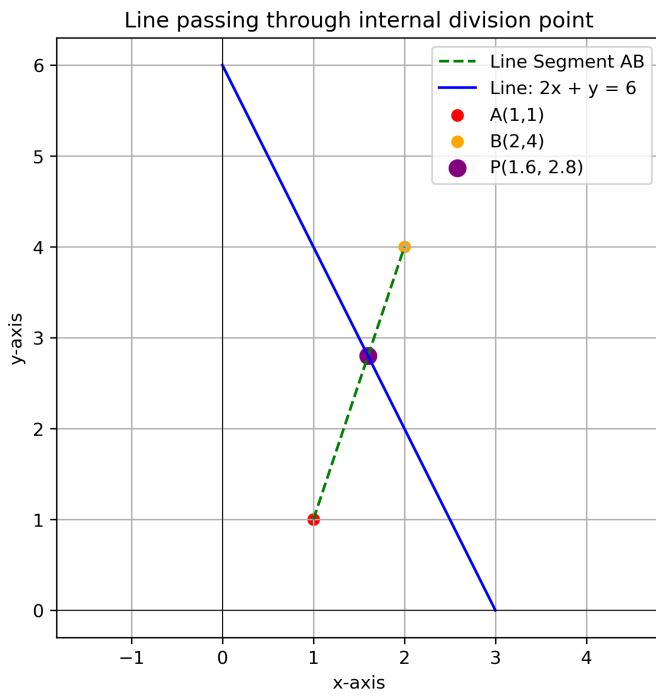


Fig. 4.1: Graph